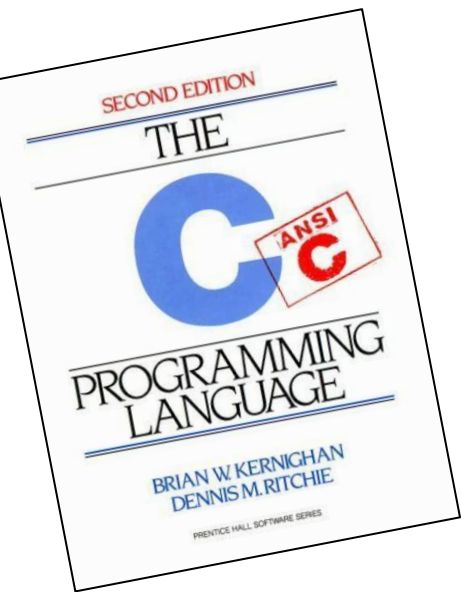
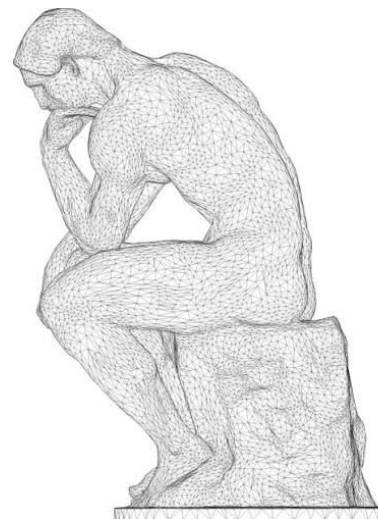




A Tutorial Introduction



Purpose

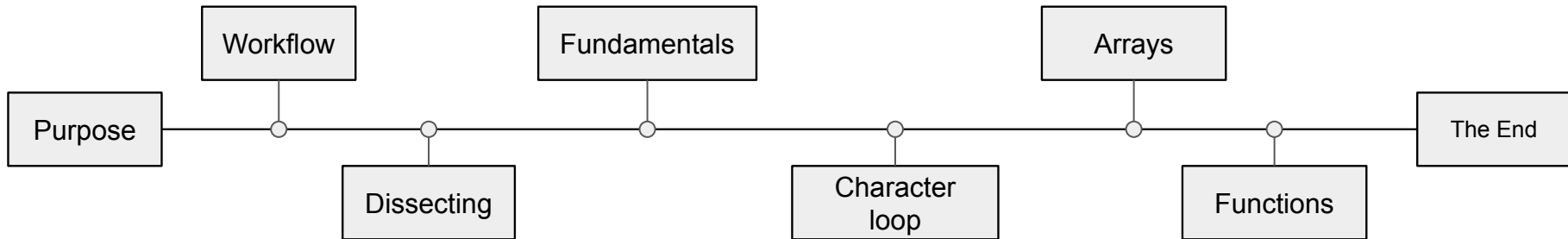
- Introduction to C
- What is C?
 - General-purpose programming language
 - Gives relatively direct access to memory and hardware
- What are we doing?
 - Get started quickly
 - Learn the rudiments
 - Mastery comes later

[illegible]

```

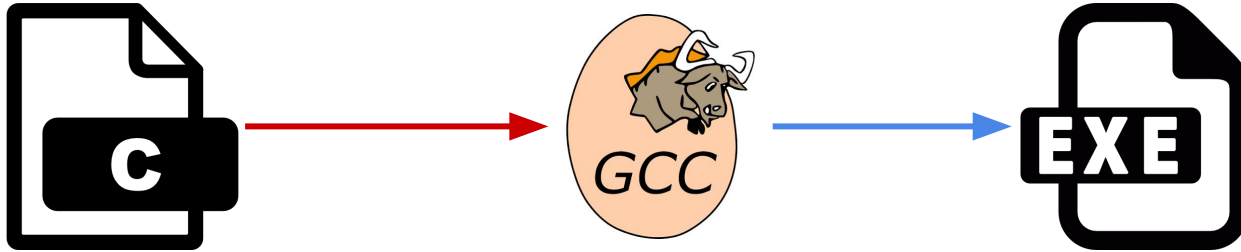
63  if (i == -1)
64      return MALL;
65
66  *p = i;
67  *p = *p + 1;
68  *p = *p + 1;
69  *p = *p + 1;
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121 *p = *p + 1;
122 *p = *p + 1;
123 *p = *p + 1;
124 *p = *p + 1;
125 *p = *p + 1;

```



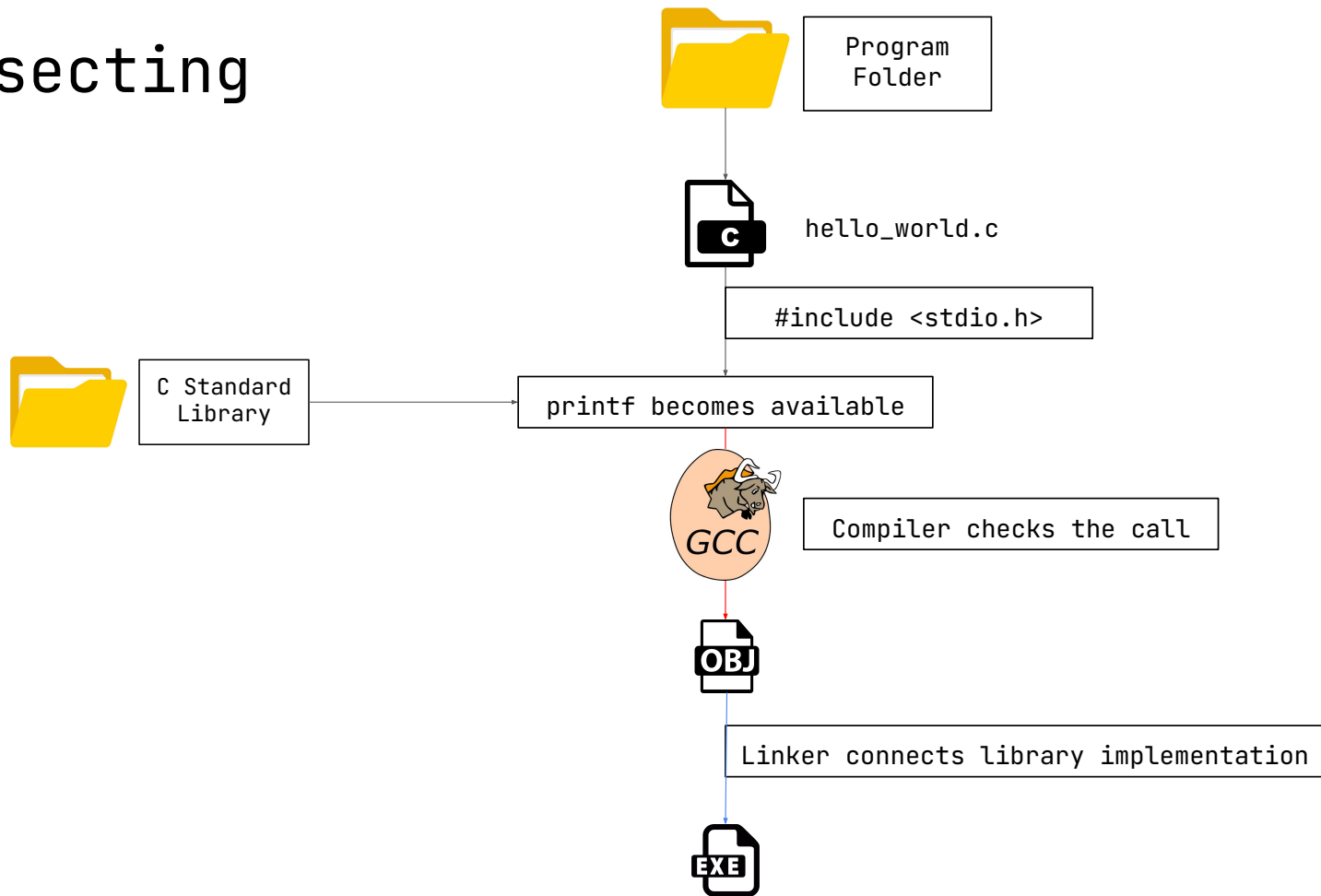
Workflow

- Write a C source file (.c)
- Compile it with a C compiler
- Run the resulting executable



This video uses GCC

Dissecting



Fundamentals

Variables store values

$$2x + 3$$

$$x = 4$$

$$2(4) + 3 = 11$$

This is similar to
variables in algebra

C can do math!

This is done through
mathematical expressions.

However, assignment '='
is done right to left.

```
x = 2 + 3;  
x is now 5
```

Loops

Loops will do repeatable
work.

It is common to increment
a counter when looping.

!Remember to be careful
with loops, as they may
go on infinitely!

Conditionals

C has conditionals to
control the program's
flow.

This is used in loops
as well.

Conditional statements
are: '=', '&&', '||'.

Character loop

Read → Check → Process → Repeat

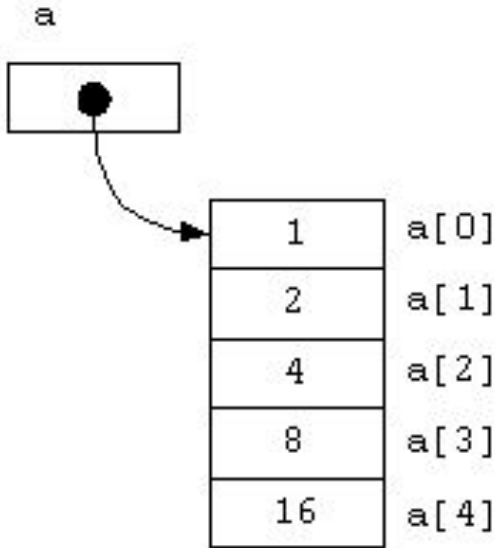
```
while((c = getchar()) != EOF)
    putchar(c);
```

EOF marks the end of the input stream.

EOF stands for End of File

Arrays

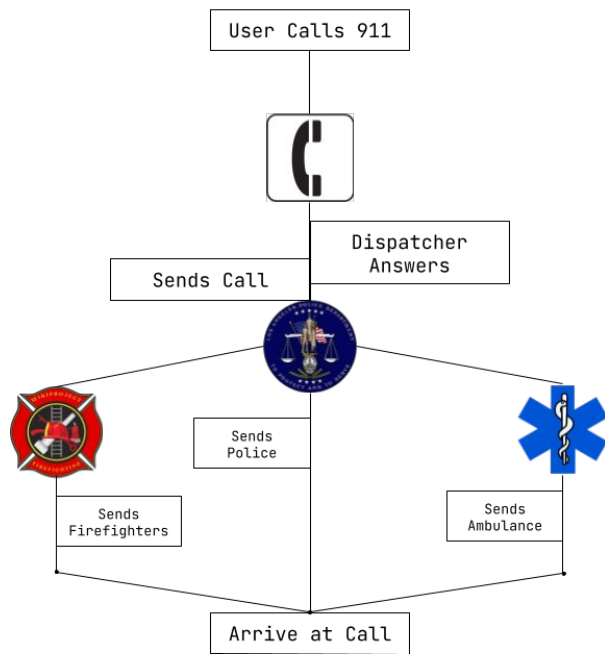
- Arrays store multiple same-type values
 - They are consecutive
 - The index starts at 0

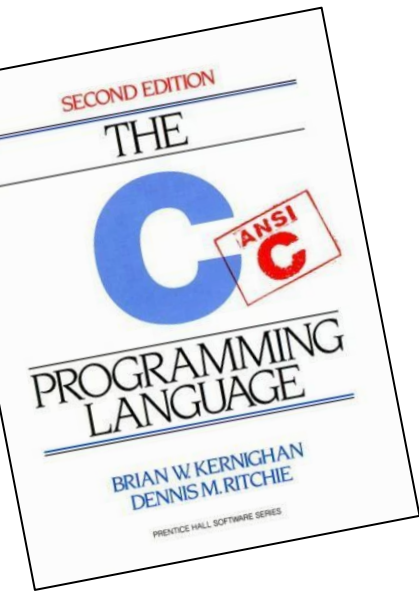


Functions

```
return-type function-name(arguments) {  
    code  
    return return-type  
}
```

```
double power(double base, int exponent)  
{  
    int i;  
    double result;  
  
    result = 1.0;  
    for(i = exponent; i > 0; i--)  
        result *= base;  
    return result;  
}
```





The first End

Chapter 1 Summary

You have seen:

- source files, compilation, and execution
- variables, expressions, and loops
- character input and output
- arrays
- functions
- Main exercise:
 - Rewrite the temperature table using a for loop.

Next:

- Types, operators, and expressions.

